



Astrophysical Benchmarks & Baseline Constants

TECHNICAL ANNEXURE DATA GUIDE

1. Fundamental Standard Metric Baseline

Constant Property Dimension	Standard Reference Value	Scientific Value (SI Wavelength)	Dataset Target Matrix Mapping
Earth Mean Radius (1 $R_{\oplus}$)	6,371.00 km	6.3710×10^6 m	Scales continuous parameter pl_rade
Earth Mass (1 $M_{\oplus}$)	5,972.20 vottaarams	5.9722×10^{24} kg	Scales continuous parameter pl_bmasse
Earth Escape Velocity ($v_{e\oplus}$)	11.186 km/s	1.1186×10^4 m/s	Ground-truth validation velocity baseline
Solar Radius (1 $R_{\odot}$)	695,700 km	6.9570×10^8 m	Scales stellar parameter st_rad
Solar Mass (1 $M_{\odot}$)	1,988,400 vottaarams	1.9884×10^{30} kg	Scales stellar parameter st_mass

2. Observed Pipeline Boundaries & Extrema

Evaluated Field Category	Maximum Measured Metric	Associated Target Planet Entity	Analytical Context & Constraints
Planetary Envelope Radius	~27.35 Jupiter Radii	Highly Inflated Gas Envelopes	Geometric max boundary checking for close-in irradiated Hot Jupiters.
Line-of-Sight Distance	8,500.00 Parsecs	Galactic Bulge Regions	Tracks coordinate column <code>sy_dist</code> (~27,722 Light Years.



3. Derived Feature Engineering Equations

3.1 Escape Velocity Model Conversion (v_e)

Calculates the exact escape velocity of any row profile using standard SI metric expansions:

$$v_e = \sqrt{(2 * G * M) / R}$$

Where $G = 6.6743 \times 10^{-11}$, $M = pl_bmasse \times 5.9722 \times 10^{24}$, $R = pl_rade \times 6.371 \times 10^6$.

3.2 Dimensionless Transit Signal Depth Calibration (pl_ratror)

Establishes geometric size validation bounds between planet disks and stellar radii:

$$pl_ratror = (pl_rade * 6.3710 \times 10^6) / (st_rad * 6.9570 \times 10^8)$$

